PRISM: Training-Free, Multimodal Failure Localization for Industrial Manipulation via Topological Persistence

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***Abstract*—State-of-the-art automatic failure-analysis pipelines for robot manipulation are predominantly vision-only: they extract keyframes from pixel motion and hand them to a vision-language model (VLM). This is a poor fit for industrial assembly, where many of the failures that matter most — contact jams, over-torqued fasteners, gripper micro-slip, controller saturation — are barely visible on camera but are unambiguous in force, torque, and proprioceptive telemetry. We present PRISM, a training-free, modality-agnostic pipeline that reframes failure localization as detecting persistent anomalies in a trajectory's raw sensor streams rather than its video alone. Per-timestep metrics from any available modality are converted into an empirical-CDF "surprise" score, aggregated across metrics, and reduced with 0-dimensional persistent homology into a small set of scale-invariant candidate events; only then are these events handed to a VLM for semantic labeling, in a stage kept structurally separate from the numeric evidence. On REASSEMBLE, a real-world, 4,551-demonstration industrial (dis)assembly benchmark with 516 labeled failures, PRISM matches or exceeds a pixel-flow baseline and a uniform-sampling baseline overall, and leads a pixel-flow baseline by up to 23 percentage points in precision on force-dominant, visually silent sub-tasks such as insertion and removal — exactly the failure class that concerns industrial deployment.**

***Index Terms*—failure detection, industrial manipulation, multimodal perception, benchmarking, vision-language models, topological data analysis**

I. Introduction

Robot learning is being asked to meet industrial standards of precision, reliability, and safety, and that bar cannot be met by systems that only report whether a rollout succeeded. Diagnosing *where* and *why* a trajectory went wrong is a prerequisite for automatic quality control, dataset curation, and teleoperator feedback in deployed manipulation systems, and doing so by hand does not scale to industrial data volumes.

Recent work automates this diagnosis, but almost exclusively from video: keyframes or scene graphs are extracted from pixel motion and a VLM or LLM is asked to explain what it sees [1]–[3], [5]. This is a structural mismatch for contact-rich industrial tasks, where the physically informative signal is often force, torque, or joint effort rather than appearance — a bolt can be visibly "in place" while its insertion telemetry shows a stall or an over-torque event. We argue that failure localization should first be treated as a *signal-detection problem* over the full sensor stack, with vision-language reasoning deferred to a later, purely semantic stage. This extended abstract focuses on validating that claim on REASSEMBLE [6], a real-world industrial assembly/disassembly benchmark, using our method, PRISM (Persistence-Ranked Identification of Salient Moments).

II. Related Work

Existing autonomous failure-analysis systems each solve part of this problem. MATT [1] localizes failure semantically (verb/argument) but has no temporal or spatial grounding and no quantitative metrics. REFLECT [3] builds hierarchical scene-graph summaries from CLIP-cropped regions and an LLM to pinpoint execution- and planning-stage failures, but relies on known object/asset states and reports success/failure rather than a graded quality signal. DQAF [4] scores teleoperated episodes against an assumed expert reference trajectory using a small set of low-level metrics, which improves interpretability for human operators but reduces generality and gives only coarse, threshold-based temporal localization. AutoEval [5] fuses raw numeric metrics and images for a VLM but does not enforce that its output stays grounded in the numbers, reports a single trajectory-level score rather than localized events, and its evaluation is largely in-distribution. KITE [2] extracts keyframes purely from pixel flow and ranks them by non-maximum suppression (NMS), a window-size hyperparameter that directly changes how many events are detected. None of these combine multiple raw modalities, a scale-invariant and parameter-light event selector, and a stage that keeps quantitative evidence from silently overriding what a VLM actually sees — the gap PRISM targets.

III. Method: PRISM

A. Multimodal metric extraction

For every trajectory, we compute a bank of per-timestep scalar signals from the modalities available in the logged data: action discontinuity, motion smoothness, joint/robot-state velocity spikes, stalling, and visual optical-flow magnitude. These signals are treated uniformly as scalar time series by the downstream surprise-scoring and event-selection stages. As a result, PRISM can degrade gracefully to the sensors present in a dataset, rather than requiring a fixed sensor suite.

B. ECDF-based surprise aggregation

For metric j with trajectory values $X_1^{(j)}, \ldots, X_n^{(j)}$, we compute left- and right-tail empirical CDFs

$$F_{\text{left}}^{(j)}(z) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{X_i^{(j)} \leq z\} \tag{1}$$

and analogously $F_{\text{right}}^{(j)}$. The per-timestep surprise score is

$$O_t^{(j)} = -\log \min\left(F_{\text{left}}^{(j)}(X_t^{(j)}),\, F_{\text{right}}^{(j)}(X_t^{(j)})\right). \tag{2}$$

Scores are summed across metrics, $O_t = \sum_j O_t^{(j)}$, and smoothed with a small-σ Gaussian filter, chosen over a box filter because a box filter can introduce spurious local extrema (ringing) that a Gaussian filter is guaranteed not to add. Add: The Salvgol filter and relu type modification We are explicit that O_t is a within-trajectory rarity heuristic, not a calibrated probability: the underlying process is neither i.i.d. nor stationary, so the Glivenko–Cantelli convergence guarantee behind the ECDF does not hold here, and summing across correlated metrics does not assume independence — it is a deliberate corroboration mechanism, since two metrics agreeing on an anomaly should reinforce it. The trade-off is that redundant metrics can inflate a peak; we treat this as a limitation mitigated by favoring modality diversity over metric count, not a correctness issue.

C. Topological event selection

Thresholded top-K peak-picking and NMS both require an arbitrary window-size hyperparameter, and both change the number of detected events as that hyperparameter changes. We instead apply 0-dimensional persistent homology to O_t: each local maximum is the birth of a connected super-level-set component, which merges into a taller neighbor at a local minimum (death); persistence is the birth height minus the merge (death) height. We rank surviving components not by raw persistence but by the *area* under the curve between each component's left/right extents Actually it is the left and right extents found by FWHM-rule. We can also switch to the left death and right death, but we are using this instead. Give reason, which down-weights instantaneous, single-timestep sensor-glitch spikes relative to sustained anomalies — the more common failure signature in robot telemetry. This forfeits the ℓ_∞ stability guarantee behind the standard "elder rule" ranking, which bounds how much small uniform vertical noise can reorder peaks; we accept this trade-off because our concern is Dirac-Delta-like single-sample spikes, not uniform noise. Unlike NMS, the number of events persistence returns is invariant to an arbitrary window choice, which we verified holds regardless of a swept persistence threshold, while NMS output count is highly sensitive to its window size.

~~Currently we are not focussing on downstream VLM eval. It is only a Key event localization. This is just a downstream use-case~~

pipeline.png

Fig. 1. PRISM pipeline: multimodal metrics → ECDF surprise scores → persistence-based, area-ranked event selection → image-only VLM labeling, aggregated with a separate metric-grounding explanation.

D. Decoupled semantic grounding

The top-K events, together with several neighboring frames (not a single frame, so temporal context such as a gripper closing is visible), are passed to a VLM that sees only images. A separate "metric-grounding" layer states in text what the underlying signals indicated at that event, and a final LLM aggregator combines the two into a per-event report. We introduced this split after observing that a VLM given both raw metric values and images will label a large metric spike as a failure without inspecting the visual evidence — decoupling the two evidence sources keeps the numeric prior from overpowering what is actually visible.

IV. Evaluation on REASSEMBLE

A. Dataset and protocol

REASSEMBLE [6] is a real-world, multimodal, contact-rich industrial assembly/disassembly dataset built on the NIST Assembly Task Board #1: 4,551 high-level demonstrations, segmented into 10,195 temporally annotated low-level action events across four task families — Pick, Insert, Remove, Place — of small industrial parts, with synchronized RGB, force/torque, audio, and proprioceptive streams. 516 high-level demonstrations are labeled failures, and every low-level event carries a start/end timestamp and a success/fail label, which lets us score localization directly against ground-truth regions rather than a single trajectory-level label. We selected REASSEMBLE specifically because it is the class of task where a purely visual pipeline is expected to struggle: failures such as an over-torqued or misaligned fastener can be visually unremarkable while unambiguous in force telemetry.

We report Recall@K (does at least one of the top-K selected events fall inside a labeled failure region?) and Precision@K

TABLE I
RECALL@K AND PRECISION@K ON REASSEMBLE ($K = 5$), BY TASK

Method	Overall	Insert	Pick	Place	Remove
Recall@K					
Uniform	0.956	0.905	1.000	1.000	1.000
KITE-based	0.656	0.460	0.900	0.910	0.740
PRISM (ours)	**0.935**	**0.880**	**0.940**	**1.000**	**0.990**
Precision@K					
Uniform	0.691	0.529	0.916	0.843	0.747
KITE-based	0.568	0.330	0.904	0.800	0.630
PRISM (ours)	**0.714**	**0.560**	**0.910**	**0.843**	**0.775**

Bold = best per column. "KITE-based" re-implements only KITE's pixel-flow + NMS keyframe front end.

(what fraction of the top-K selected events fall inside a labeled failure region?), using the same evidence budget $K = 5$ used throughout our broader evaluation. We compare against Uniform sampling and a KITE-based baseline; because the question here is specifically about event selection, we re-implement only KITE's pixel-flow-plus-NMS temporal front end and not its downstream pseudo-bird's-eye-view spatial reasoning, isolating the selection mechanism rather than the full KITE system. Results were checked for consistency across multiple camera viewpoints in the dataset.

B. Results

Overall Recall@K is high for every method, including Uniform (0.956), because failure regions in REASSEMBLE are wide: a labeled failure event occupies, on average, 69.8% of the duration of its failure trajectory, so almost any sampling strategy places at least one event inside it. Recall@K is therefore weakly discriminative on this benchmark, which is why we treat Precision@K as the primary statistic here and add a point-level check in Section IV-C. PRISM attains the best overall precision (0.714 vs. 0.691 for Uniform and 0.568 for KITE-based).

More importantly for an industrial audience, the gap between PRISM and the vision-only baseline is largest exactly where vision is the wrong sensor. On Insert — threading, seating, and contact events dominated by force cues that a flow-based detector cannot see — PRISM's precision exceeds KITE-based by 23 percentage points (0.560 vs. 0.330). On Remove the gap is 14.5 points (0.775 vs. 0.630). On Pick and Place, where the object visibly moves, all three methods converge because pixel flow is already a reasonably strong signal there. This is the central empirical claim of this abstract: telemetry-grounded event selection recovers precisely the failure class that a purely visual pipeline is structurally unable to see, without any task-specific training, and the result is stable across camera viewpoints.

C. Point-level validation

Because REASSEMBLE's wide failure windows compress how discriminative Recall@K/Precision@K can be, we additionally evaluate on RoboFail [3], 30 real-world trajectories annotated with a single failure timestamp rather than a region, using the median absolute error between the nearest selected event and the annotated failure point. PRISM reaches 5.80 s median error versus 12.87 s (KITE-based) and 13.03 s (Uniform) — more than a $2\times$ reduction relative to both baselines — which confirms the REASSEMBLE advantage is not an artifact of generous ground-truth windows but reflects genuinely tighter localization.

V. DISCUSSION AND LIMITATIONS

PRISM's aggregation sums per-metric surprise scores, so redundant metrics that measure near-identical quantities can jointly inflate a single peak; we treat corroboration between independent evidence sources as a feature but acknowledge this as a limitation, best mitigated by favoring modality diversity over metric count rather than by any change to the scoring rule. Ranking by area rather than raw persistence trades away the ℓ_∞ stability guarantee of the standard "elder rule"; we accept this because our target failure mode — single-timestep sensor spikes — is exactly the case that guarantee does not protect against. needs confirmation Finally, the 5.80 s residual on RoboFail is not zero: qualitatively, PRISM's selected events sometimes precede or follow the visually confirmed failure onset because a physical cause (e.g. a slip) and its externally logged or visible consequence are not simultaneous; characterizing this lag, and replacing our hand-set event-merging weights with values swept across many trajectories rather than chosen by hand, are open problems we are addressing directly in response to internal review.

VI. CONCLUSION

PRISM treats failure localization for industrial manipulation as a training-free, modality-agnostic signal-detection problem: topological persistence replaces arbitrary top-K or NMS peak-picking, and a decoupled metric/vision architecture keeps VLM semantic reasoning from being overridden by raw numbers. On REASSEMBLE, PRISM matches strong baselines on visually obvious task families and clearly outperforms a vision-only baseline precisely on the force-dominant, visually silent failures — insertion and removal — that most concern industrial deployment. We take this as evidence for a broader claim: closing the reliability gap between lab demonstrations and industrial deployment requires treating the full sensor stack, not the camera alone, as primary evidence for what went wrong.

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